



Dement's Sleep & Dreams:

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Stanford
M E D I C I N E

**Why do we sleep?
Why do we dream?
Do our dreams have meaning?**



Questions

Have you ever seen a classmate fall asleep in class?

Have you ever fallen asleep doing homework?

Do you wake up tired?

Have you ever skipped sleep to do your homework?

Have you ever seen a person drive drowsy?

Questions

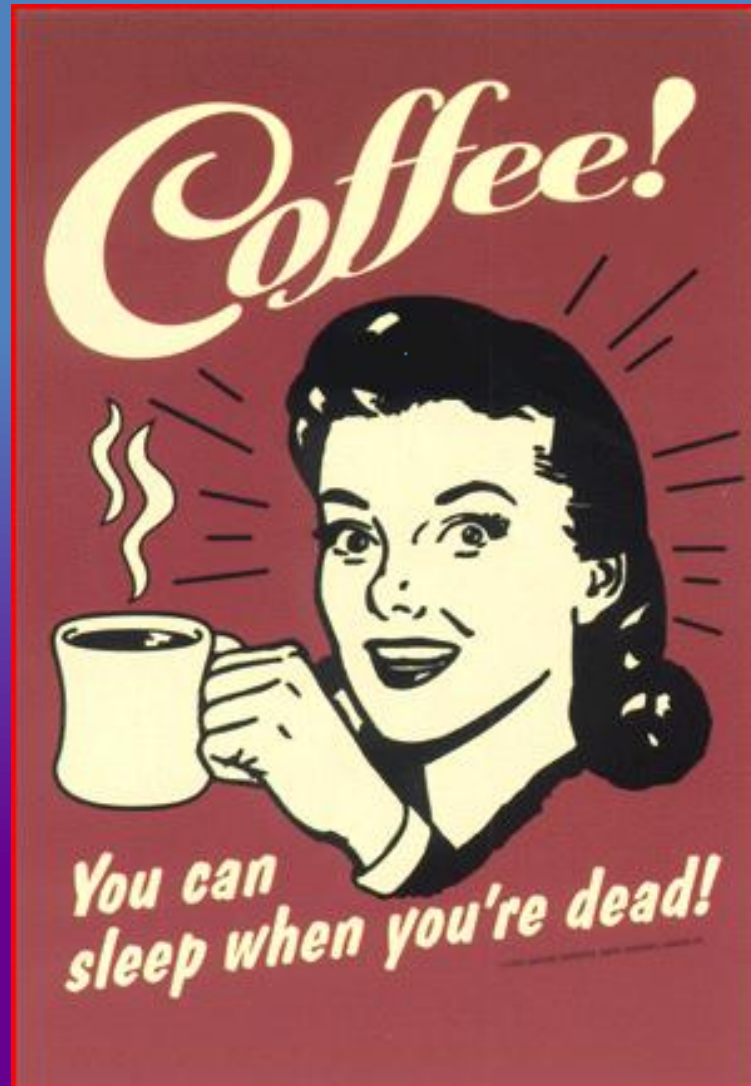
In high school did you get to sleep after your parents did?

Do you get more sleep now than in high school?

Are you ever been woken up by a text message?

Do you wish you could have more sleep?

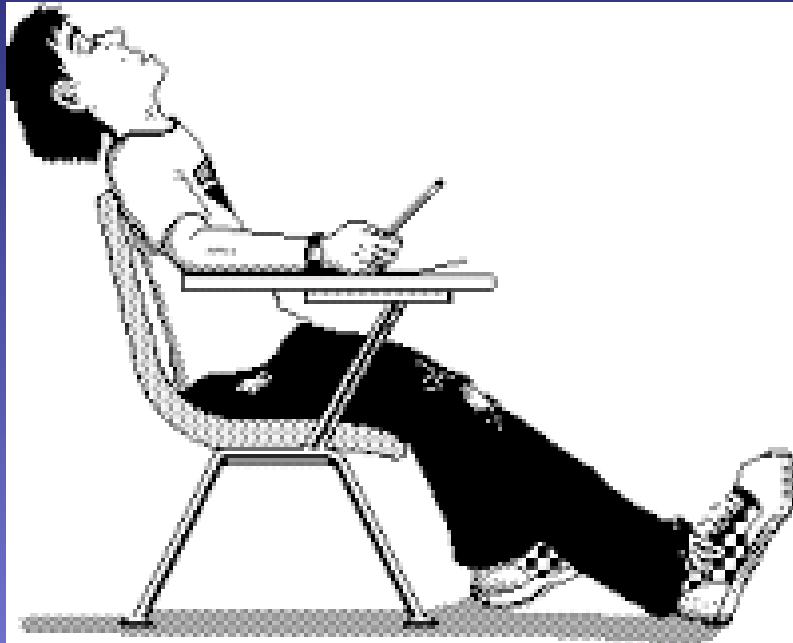
If you did not have to sleep,



would you?

Topics

- Normal sleep
- What can go wrong
- How to fix it





Drowsiness is **RED** Alert !



Housekeeping

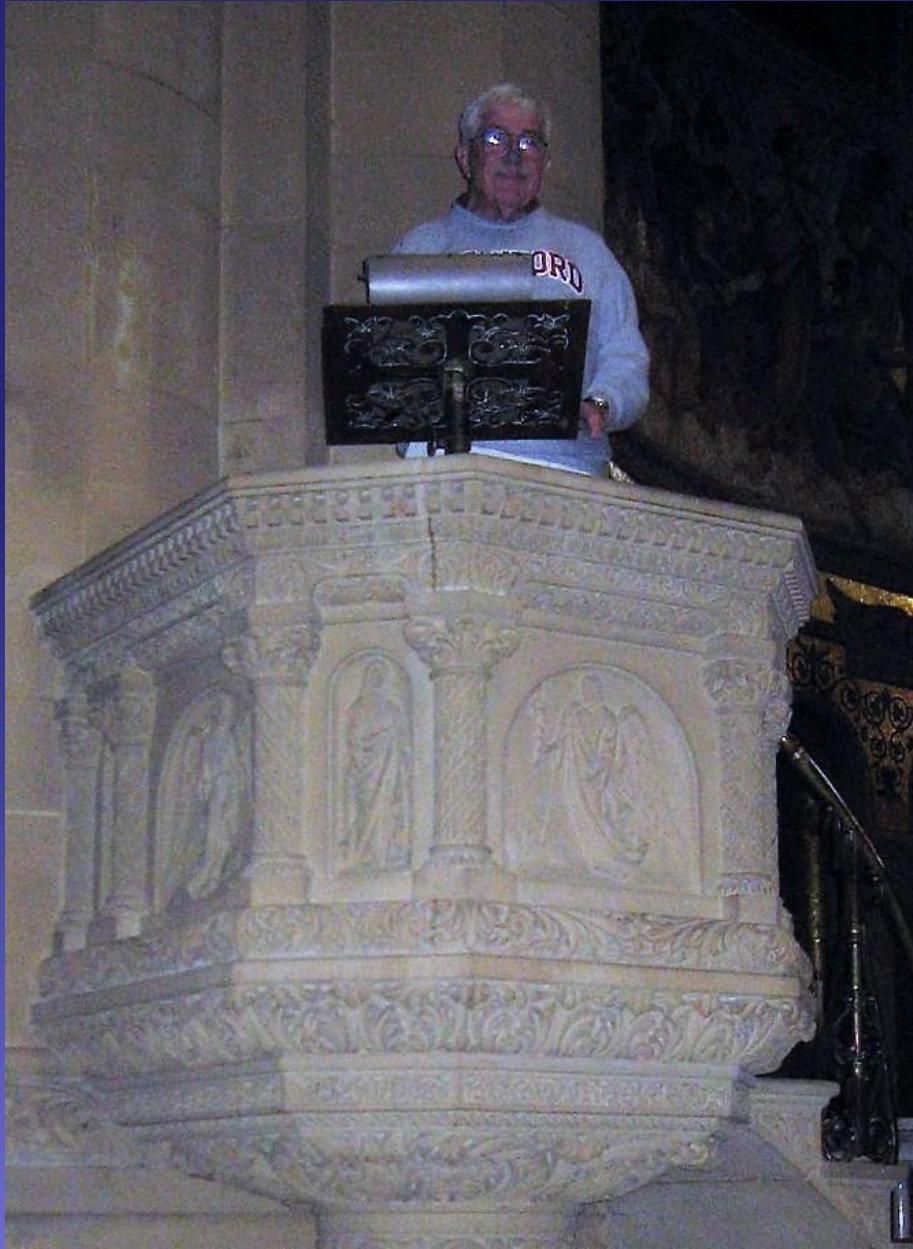
- Structure of the course
- Undergrads PYSC 135 and Grad students PSYC 235
- Sleep Journal
- Attendance!
- Extensions
- Bonus points
- T-Shirts
- Welcome DCI fellows
- Readings
- EXAMS
- **We need a volunteer for Wednesday!**

Please connect with your TA's

**You are now part of a Stanford tradition that
dates back to 1971**



**Please welcome Dr. Dement's Daughter
Mrs. Cathy Roos**



First class held
in Memorial
Church in 1971
because no
other space was
available

Clinical Sleep Medicine was first successfully established at Stanford in 1970

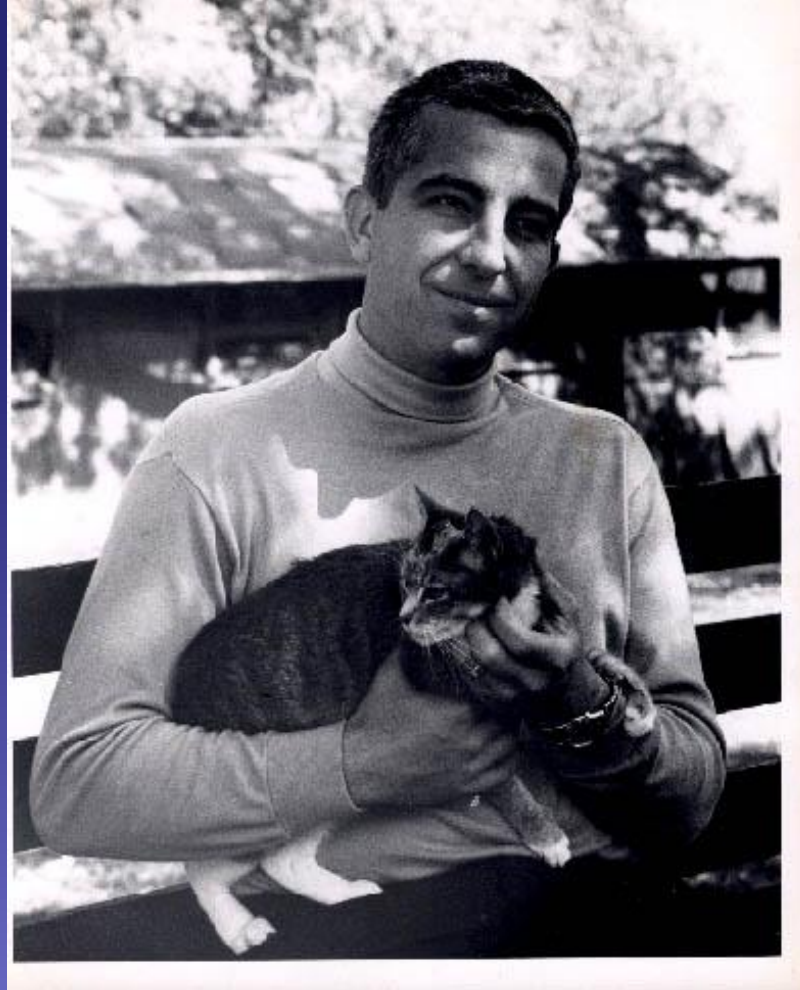


William C. Dement
1928-2020



Christian Guilleminault
1938-2019

Dement created Sleep & Dreams in 1971



<https://www.youtube.com/watch?v=UWkeI45kbNU>

Association of Sleep Disorders Centers



hereby acknowledges that

WILLIAM C. DEMENT, M.D., PH.D.

*has satisfied the Association's
technical, clinical, and scientific standards
and is certified as a*

CLINICAL POLYSOMNOGRAPHER

*to evaluate and diagnose
sleep disorders and sleep-related disorders.*

APRIL 6, 1978

William Dement
President

Merill M. Mitler
Secretary-Treasurer

Thomas Roth
Chairperson, Committee on
Clinical Polysomnography



STANFORD UNIVERSITY MEDICAL CENTER

STANFORD, CALIFORNIA 94305 • (415) 321-1200

STANFORD UNIVERSITY SCHOOL OF MEDICINE
Sleep Disorders Clinic and Laboratory

COPY

New Patients:

The first visit to the Sleep Disorders Clinic will cost \$50.00 for this includes an intake interview and a thorough evaluation of your sleep disorder by the Director of the Clinic, William C. Dement, M.D., Ph.D. Each visit with Dr. Dement after the first will cost \$15.00. Daytime diagnostic electroencephalographic sleep polygraphy recordings range from \$50.00 to 100.00 depending on the amount of time spent, number of technicians needed etc. If all-night diagnostic EEG sleep recordings are necessary, they usually run around \$150.00, but again, this depends on amount of time spent in the laboratory, etc.

“Sleep” lost over GER mishap

By **AN LE NGUYEN**
SENIOR STAFF WRITER

Prof. William Dement’s popular “Sleep and Dreams” (PSYC 135) course was stripped of its natural science general education requirement (GER) because of a breakdown in the GER approval process. A change in the process required School of Medicine Dean Philip Pizzo to petition for the course to retain its GER status; when Pizzo failed to petition, the course lost its GER. But in an administrative error, the Registrar mistakenly listed “Sleep and Dreams” as fulfilling the natural science GER for this quarter and allowed it to retain the status.

Dement told The Daily that the University has not informed him about any of these complications, which led to the confusion and controversy surrounding “Sleep

Sciences and the School of Engineering due to their close involvement with the undergraduate curriculum.

As indicated by Gelber, procedural hurdles have emerged because “Sleep and Dreams” is taught in the psychiatry department, which falls under the School of Medicine.

“[PYSC 135] is not a GER, the only way that has not happened,” Gelber explained.

Dement told The committee coordinated the Senate’s “Sleep and Dreams” in an effort.

“The Subcommittee on the S-GER last year

2 ♦ Thursday, February 5, 2009

NEWS

Dement pushes for GER

Students think “Sleep” should satisfy natural science GER

By **AN LE NGUYEN**
SENIOR STAFF WRITER

Professor William Dement initiated a large-scale mail campaign on Tuesday, urging students to show the University that they would not be likely to enroll in his perennially popular “Sleep and Dreams” (PSYC 135) course if it no longer fulfilled the natural science general education requirement (GER).

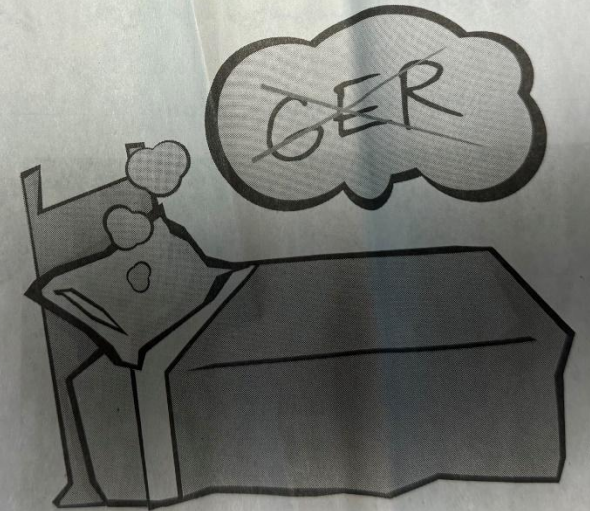
Dement told students to vote in an online poll hosted by The Daily to make their opinions known. He also told students in an email that if at least 1000 people responded to the poll, every student in the class would get bonus points. When the poll closed at 2:12 am, 805 people had voted and 679 had followed Dement’s recommendation, indicating that not having the GER status might be a deal-breaker.

Dement told The Daily that he was very happy with the online poll but also acknowledged the fact that it introduced some complexity to the issue. He hopes to present the poll results as part of the GER application for “Sleep and Dreams” when he meets with the Senate committee on GERs in late spring.

In the near future, he recognizes there will come a day when he will no longer teach the course.

“I don’t know if I’ll be a professor,” he said. “It isn’t clear

something that is hard to find in other places. Rheezen agreed. “Sleep, sleep health and sleep science in the realm of natural science in some way or form. I do



CRIS BALITISTA/
The Stanford Daily

In Winter 2009, this course lost General Education Requirement

It was reinstituted with the new
WAYS curriculum

Sleep is a natural restorative physiological process that is characterized by perceptual disengagement and must be rapidly reversible

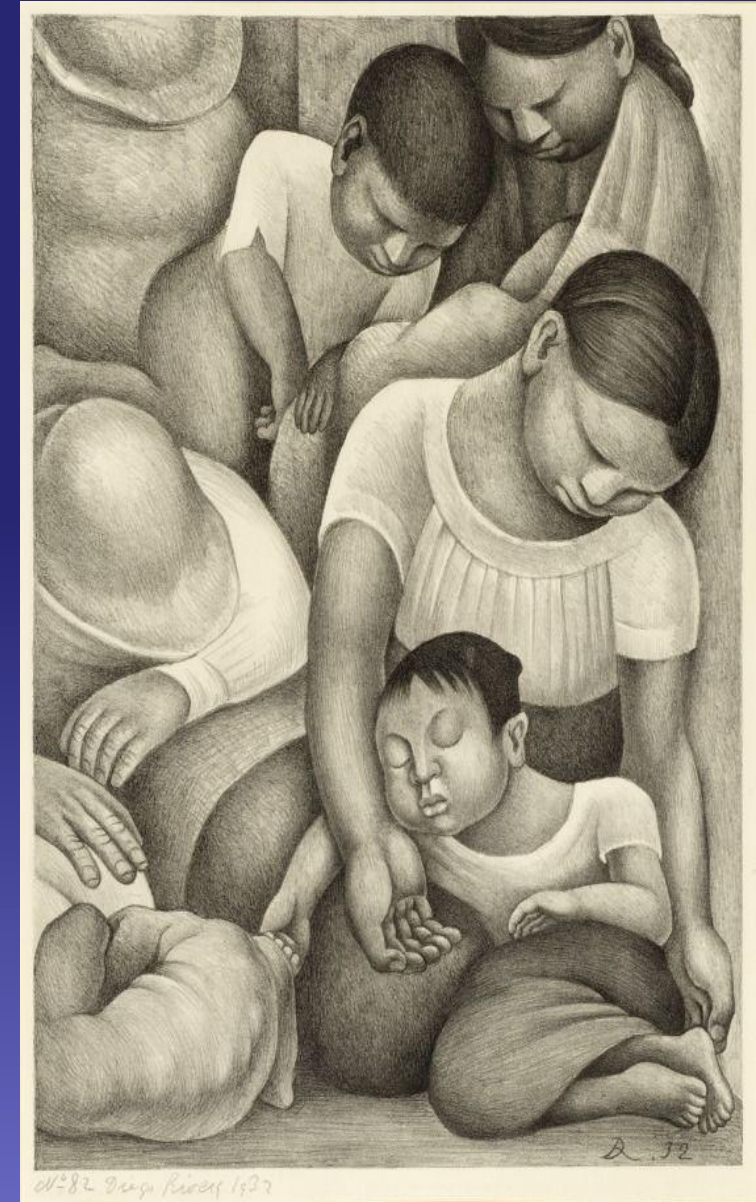


“Dream to Remember, Dream to Forget”

Sleep is a paradox



Sleep Is The Most Natural And Powerful Form Of Self-care For All Of Us



Sueño by Diego
Rivera

Functions of sleep

- Higher-order cognitive function involving mechanisms of learning and memory and synaptic plasticity



- Restorative function including maintenance of brain energy metabolism, macromolecular biosynthesis, and removal of metabolic waste

Dream Characteristics

- Intense emotions
- Hard to remember afterwards
- Illogical/ impossible situation not questioned
- Odd/old memories incorporated into content
- Fascinating to the dreamer

Early Dreams and Society

- How would a prehistoric society explain why they could see their departed loved ones in a dream?
- The need to explain dreams may be the origin of early religions
- A great leader is said to have a “vision”. That vision will unite the people
- Inspiration of dreams may have led to the establishment of great societies

Why would *Homo naledi* bury their dead?

The ancestor we never knew

The discovery of a human ancestor, *Homo naledi*, in an ancient burial chamber in South Africa has been described by scientists as the most significant find in the study of human evolution for four decades.

Timeline

Where *Homo naledi* could be dated

Today

Homo naledi

A. reidelbergensis
A. denisovaensis

One million years ago

A. erectus

Two million

A. habilis
A. rudolfensis
A. africanus
A. garhi

Three million

A. boisei
A. afarensis
A. anamensis

Four million

A. anamensis

Homo naledi

Homo naledi

Using bone scans, clay and liberal amounts of beer for an artist has recreated the face of *Homo naledi*

Skull
Was around half the size of ours at 560 cubic centimetres

Human skull

Rising Star Cave
So named because it is so tight you have to put one arm above your head

Turkana Boy
Homo erectus
1500 years ago
Advanced male
SH 50 52kg

"Lucy"
Australopithecus afarensis
3.2m years ago
Adult female
SH 64, 27-29kg

Carved fingers
An ape-like adaptation, useful in gripping them

Humanlike feet
Designed for standing upright and walking long distances

The caves
The Dinaledi chamber containing the bones translates, in local dialect, to "Chamber of Stars"

Rising star cave

Entrance

Sapien's cave
So named because it is so tight you have to put one arm above your head

Dinaledi chamber

Dragon's Back

Fossil site

AFRICA

ETHIOPIA
Afar Depression
Discovery of LUCY reveals (1974)

KENYA
Lake Turkana
Discovery of Turkana Boy (1984)

Rising star cave

Mozambique

Johannesburg

SOUTH AFRICA

Above right: inside the Rising Star cave.
Below: an artist's impression of an *Homo naledi* group depending on one of their own in the cave

Dreams and Religion

- Most if not all religions discuss dreams
- Dreams are a source of Divine Revelations
- In the book of Genesis, God must rest and Adam is put to sleep to make Eve



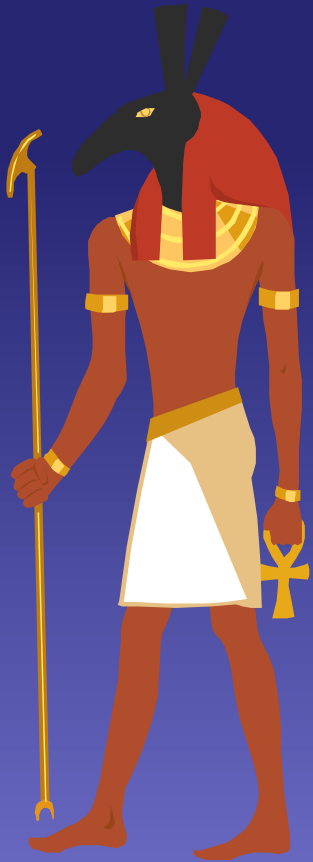
**Sumerians recorded dreams over
5,000 years ago on clay tablets**



Ancient Egypt

Dreams viewed as a guiding force and as a journey to the underworld

Special dream beds inside the temples



Hinduism



Dreams are real and caused by
the Supreme Brahman



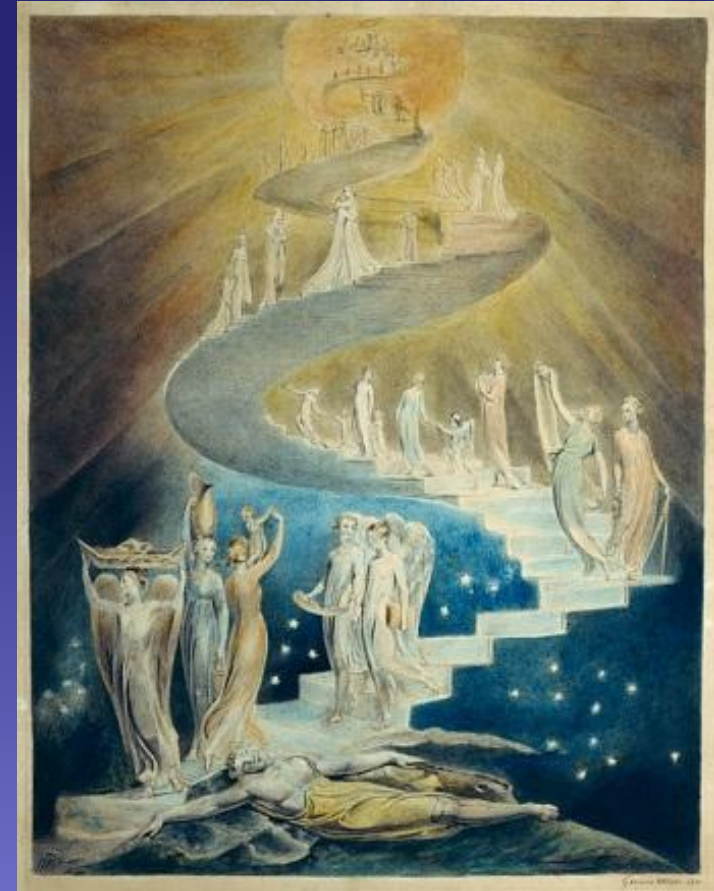


Queen Maya
dreams her son
will be the
Enlighten One



Old Testament

- Jacob's dream of the Divine ladder at Bethel (Gen 28:11-19):



New Testament



And Joseph her husband, being a righteous man and not wanting to disgrace her, planned to send her away secretly. But when he had considered this, behold, an angel of the Lord appeared to him in a dream, saying, "Joseph, son of David, do not be afraid to take Mary as your wife; for the Child who has been conceived in her is of the Holy Spirit.

Matthew 1:20

The Night Journey and Islam



The Prophet Mohammed travels to Jerusalem and Heaven

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California Child Who Died Was Allegedly Subject to Exorcism

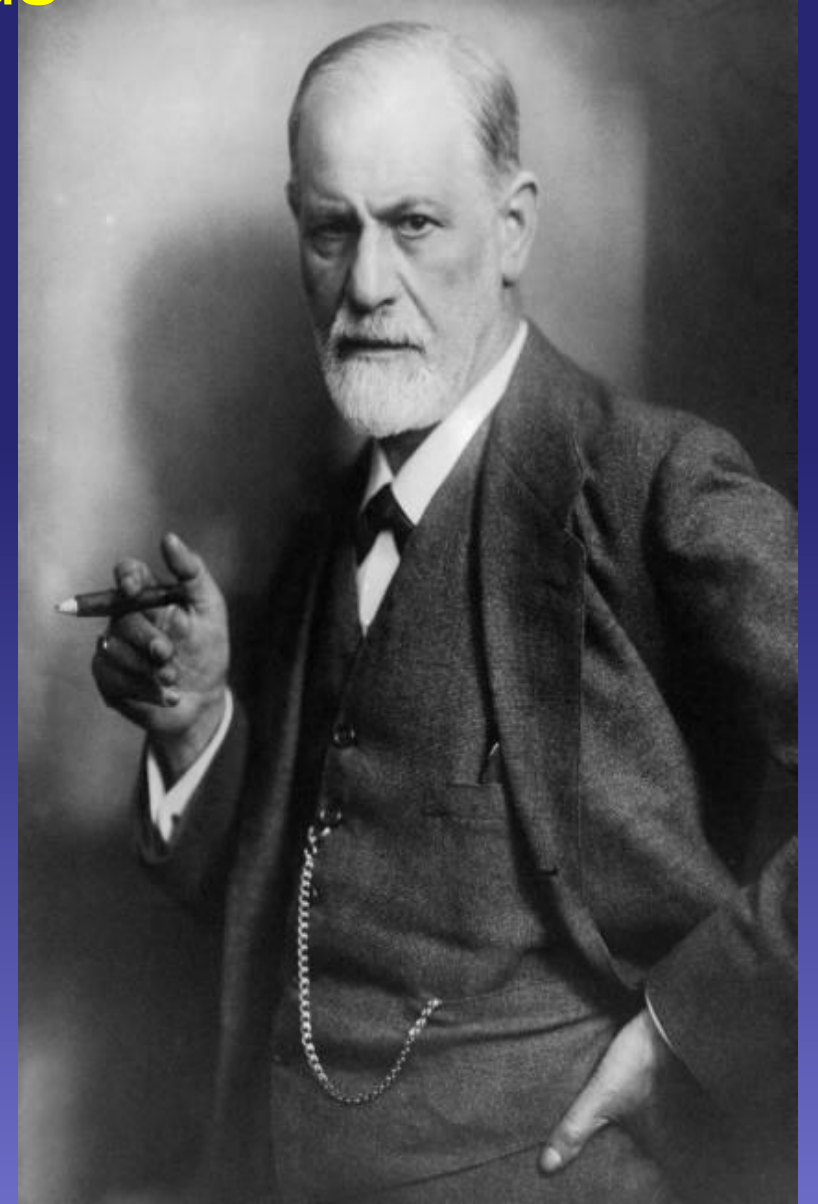
A 3-year-old girl whose death last fall has been ruled a homicide was the subject of an alleged exorcism last year at a Northern California church.

By [Associated Press](#) | May 10, 2022, at 7:38 p.m.

San Jose, California 2022

Royal Road To The Unconscious

- *Interpretation of Dreams* Freud 1899
- The content of dreams is driven by unconscious wish fulfillment. Sexual focus probably reflected the cultural repression. His focused evolved
- 25 years afterwards the human EEG recorded by Hans Berger in 1924



THE PHENOMENA OF SLEEP.



THE cry for sleep is ever greater than the cry for bread. Existence depends on both; but we eat to sleep, while we sleep to live. Sleep is of far greater importance than food for the preservation of life.

Sleep is the chiefest thing in physic. It has neither substitute nor rival. Take away sleep, hope even is gone, nothing is left.

*The
chiefest
thing*

The Witchery of Sleep by Willard Moyer



Sweet Dreams to all!
See you Wednesday